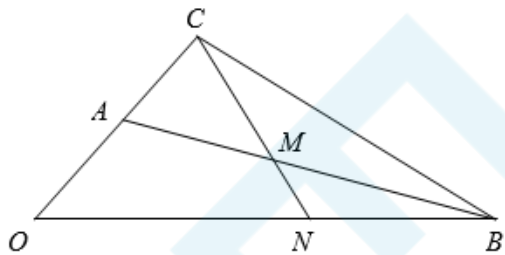


Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
			
(a)	$\overrightarrow{CM} = \overrightarrow{CA} + \overrightarrow{AM} = \overrightarrow{CA} + \frac{1}{2} \overrightarrow{AB}$	M1	This mark is given for a method to find an expression for $\overrightarrow{CM}$
	$\overrightarrow{CM} = -\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = -\frac{3}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$	A1	This mark is given for a correct expression for $\overrightarrow{CM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
(b)	$\overrightarrow{ON} = \overrightarrow{OC} + \overrightarrow{CN} = \overrightarrow{OC} + \lambda \overrightarrow{CM}$	M1	This mark is given for a method to find an expression for $\overrightarrow{ON}$
	$\overrightarrow{ON} = 2\mathbf{a} + \lambda \left(-\frac{3}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} \right)$ $= \left(2 - \frac{3}{2}\lambda \right) \mathbf{a} + \left(\frac{1}{2}\lambda \right) \mathbf{b}$	A1	This mark is given for a correct expression for $\overrightarrow{ON}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
(c)	$\left(2 - \frac{3}{2}\lambda \right) = 0 \text{ so } \lambda = \frac{4}{3}$	M1	This mark is given for deducing that the coefficient of $\mathbf{a} = 0$ and finding a value for λ
	$\overrightarrow{ON} = 0 \times \mathbf{a} + \frac{2}{3} \mathbf{b}$ Hence $ON:NB = \frac{2}{3} : \frac{1}{3} = 2:1$	A1	This mark is given for finding $\overrightarrow{ON}$ and giving a valid conclusion
(Total 6 marks)			

Q2.

Question	Scheme	Marks	AOs
	$\vec{OA} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$, $\vec{OB} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, $\vec{OC} = a\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, $a < 0$ $\vec{AB} = \vec{BD}$, $ \vec{AB} = 4$		
(a)	E.g. $\vec{OD} = \vec{OB} + \vec{BD} = \vec{OB} + \vec{AB}$ or $\vec{OD} = \vec{OB} + \vec{BD} = \vec{OB} + \vec{AB} = \vec{OB} + \vec{OB} - \vec{OA} = 2\vec{OB} - \vec{OA}$ or $\vec{OD} = \vec{OB} + \vec{BD} = \vec{OB} + \vec{AB} = \vec{OA} + \vec{AB} + \vec{AB} = \vec{OA} + 2\vec{AB}$		
	$= \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ -5 \\ 7 \end{pmatrix}$ or $= \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + 2 \left(\begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -5 \\ 7 \end{pmatrix}$	M1	3.1a
	$= \begin{pmatrix} 6 \\ -7 \\ 10 \end{pmatrix} \text{ or } 6\mathbf{i} - 7\mathbf{j} + 10\mathbf{k}$	A1	1.1b
		(2)	
(b)	$(a-2)^2 + (5-3)^2 + (-2-4)^2$	M1	1.1b
	$\left\{ \begin{array}{l} \vec{AC} = 4 \Rightarrow (a-2)^2 + (5-3)^2 + (-2-4)^2 = 4^2 \\ \Rightarrow (a-2)^2 = 8 \Rightarrow a = \dots \text{ or } \Rightarrow a^2 - 4a - 4 = 0 \Rightarrow a = \dots \end{array} \right.$	dM1	2.1
	$(\text{as } a < 0 \Rightarrow) a = 2 - 2\sqrt{2} \text{ (or } a = 2 - \sqrt{8})$	A1	1.1b
		(3)	

(5 marks)

Notes for Question	
(a)	
M1:	Complete <i>applied</i> strategy to find a vector expression for $\vec{OD}$
A1:	See scheme
Note:	Give M0 for subtracting the wrong way wrong to give e.g. $(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) + (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}) - (4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) = (4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) + (-2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}) = (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k})$
Note:	Writing e.g. $\vec{OD} = \vec{OB} + \vec{AB}$ or $\vec{OD} = 2\vec{OB} - \vec{OA}$ with no other work is M0
Note:	Finding <i>coordinates</i> , i.e. $(6, -7, 10)$ without reference to the correct position vectors is A0
Note:	Allow M1A1 for writing down $6\mathbf{i} - 7\mathbf{j} + 10\mathbf{k}$ with no working
Note:	M1 can be implied for at least two correct components in their position vector of D
(b)	
M1:	Finds the difference between $\vec{OA}$ and $\vec{OC}$, then squares and adds each of the 3 components Note: Ignore labelling
dM1:	Complete method of <i>correctly</i> applying Pythagoras' Theorem on $ \vec{AC} = 4$ and using a correct method of solving their resulting quadratic equation to find at least one of $a = \dots$
Note:	Condone at least one of either awrt 4.8 or awrt -0.83 for the dM mark
A1:	Obtains only one exact value, $a = 2 - 2\sqrt{2}$
Note:	Writing $a = 2 \pm 2\sqrt{2}$, without evidence of rejecting $a = 2 + 2\sqrt{2}$ is A0
Note:	Allow exact alternatives such as $2 - \sqrt{8}$ or $\frac{4 - \sqrt{32}}{2}$ for A1, and isw can be applied
Note:	Writing $a = -0.828\dots$, without reference to a correct exact value is A0

Q3.

Question	Scheme	Marks	AOs
(a)	Attempts two of the relevant vectors $\pm \overrightarrow{AB} = \pm(-4\mathbf{i} + 7\mathbf{j} + \mathbf{k})$ $\pm \overrightarrow{AC} = \pm(-20\mathbf{i} + (p+3)\mathbf{j} + 5\mathbf{k})$ $\pm \overrightarrow{BC} = \pm(-16\mathbf{i} + (p-4)\mathbf{j} + 4\mathbf{k})$	M1	3.1a
	Uses two of the three vectors in such a way as to find the value of p . E.g. $p+3 = 5 \times 7$	dM1	2.1
	$p = 32$	A1	1.1b
		(3)	
(a) Alternative:			
	$r_{AB} = 4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k} + \lambda(-4\mathbf{i} + 7\mathbf{j} + \mathbf{k})$	M1	3.1a
	$4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k} + \lambda(-4\mathbf{i} + 7\mathbf{j} + \mathbf{k}) = -16\mathbf{i} + p\mathbf{j} + 10\mathbf{k} \Rightarrow \lambda = 5$ $4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k} + \lambda(-4\mathbf{i} + 7\mathbf{j} + \mathbf{k}) = -16\mathbf{i} + p\mathbf{j} + 10\mathbf{k} \Rightarrow p = 35 - 3$	dM1	2.1
	$p = 32$	A1	1.1b
(b)	Deduces that $\overrightarrow{OD} = \lambda \overrightarrow{OB} = 4\lambda\mathbf{j} + 6\lambda\mathbf{k}$ and attempts $\overrightarrow{CD} = 16\mathbf{i} + (4\lambda - 32)\mathbf{j} + (6\lambda - 10)\mathbf{k}$	M1	3.1a
	Correct attempt at λ using the fact that $\overrightarrow{CD}$ is parallel to $\overrightarrow{OA}$ $\overrightarrow{CD} = 16\mathbf{i} + (4\lambda - 32)\mathbf{j} + (6\lambda - 10)\mathbf{k}$ $\overrightarrow{OA} = 4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ $4\lambda - 32 = -12 \Rightarrow \lambda = \dots$ OR $6\lambda - 10 = 20 \Rightarrow \lambda = \dots$	dM1	1.1b
	$ \overrightarrow{OD} = 5 \times \sqrt{4^2 + 6^2} = 10\sqrt{13}$	A1	1.1b
		(3)	
(b) Alternative:			
	Deduces that $\overrightarrow{OD} = \lambda \overrightarrow{OB} = 4\lambda\mathbf{j} + 6\lambda\mathbf{k}$ and attempts $\overrightarrow{OD} = \overrightarrow{OC} + \mu \overrightarrow{OA} = -16\mathbf{i} + 32\mathbf{j} + 10\mathbf{k} + \mu(4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k})$	M1	3.1a
	Correct attempt at λ or μ using the fact that $\lambda \overrightarrow{OB} = \overrightarrow{OC} + \mu \overrightarrow{OA}$ E.g. $-16 + 4\mu = 0 \Rightarrow \mu = 4$	dM1	1.1b
	$ \overrightarrow{OD} = 5 \times \sqrt{4^2 + 6^2} = 10\sqrt{13}$	A1	1.1b
		(3)	
(6 marks)			
Notes:			

(a)

M1: Attempts two of the three relevant vectors by **subtracting** either way around. See scheme.

Allow equivalent work e.g. $\pm \overrightarrow{AB} = \pm (\overrightarrow{OB} + \overrightarrow{AO})$

If no working is shown, method can be implied by 2 correct components.

dM1: For the key step in using the fact that if the vectors are parallel, they will be multiples of each other (where the multiple is something other than 1) to find p .

E.g. $p+3=5 \times 7$, $p-4=\frac{4}{5}(p+3)$, $p-4=4 \times 7$

A1: $p=32$ (Condone 32j)

For reference, $\overrightarrow{BC} = 4\overrightarrow{AB}$, $\overrightarrow{AC} = 5\overrightarrow{AB}$, $\overrightarrow{BC} = \frac{4}{5}\overrightarrow{AC}$, $\overrightarrow{AC} = \frac{5}{4}\overrightarrow{BC}$

Note that candidates generally only need to use 2 components to find p and if the 3rd component has errors but is not used, full marks can be awarded.

Alternative:

M1: Forms the vector equation using A or B as position and $\pm \overrightarrow{AB}$ as the direction

dM1: For the key step in using the fact that C lies on the line to find p

A1: $p=32$ (Condone 32j)

For reference, $\overrightarrow{BC} = 4\overrightarrow{AB}$, $\overrightarrow{AC} = 5\overrightarrow{AB}$, $\overrightarrow{BC} = \frac{4}{5}\overrightarrow{AC}$, $\overrightarrow{AC} = \frac{5}{4}\overrightarrow{BC}$

Note that candidates generally only need to use 2 components to find p and if the 3rd component has errors but is not used, full marks can be awarded.

There will be other approaches e.g. using “gradients” and “ratios” and the method marks can be implied – if you are unsure if such attempts deserve credit use Review

(b) Vector approach

M1: Deduces that $\overrightarrow{OD} = \lambda \overrightarrow{OB} = 4\lambda \mathbf{j} + 6\lambda \mathbf{k}$ and attempts $\overrightarrow{CD} = 16\mathbf{i} + (4\lambda - 32)\mathbf{j} + (6\lambda - 10)\mathbf{k}$

dM1: Correct attempt at finding λ using the fact that $\overrightarrow{CD}$ is parallel to $\overrightarrow{OA}$

E.g. $16\mathbf{i} + (4\lambda - 32)\mathbf{j} + (6\lambda - 10)\mathbf{k} = 4\alpha\mathbf{i} - 3\alpha\mathbf{j} + 5\alpha\mathbf{k} \Rightarrow \alpha = 4 \Rightarrow 4\lambda - 32 = -3 \times 4 \Rightarrow \lambda = \dots$

A1: $|\overrightarrow{OD}| = 10\sqrt{13}$

Alternative:

M1: Deduces that $\overrightarrow{OD} = \lambda \overrightarrow{OB} = 4\lambda \mathbf{j} + 6\lambda \mathbf{k}$ and attempts

$$\overrightarrow{OD} = \overrightarrow{OC} + \mu \overrightarrow{OA} = -16\mathbf{i} + 32\mathbf{j} + 10\mathbf{k} + \mu(4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k})$$

dM1: Correct attempt at finding λ or μ using the fact that $\lambda \overrightarrow{OB} = \overrightarrow{OC} + \mu \overrightarrow{OA}$

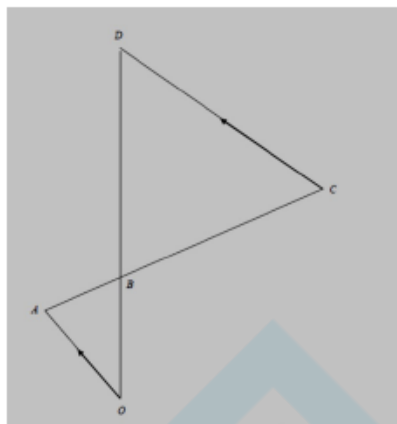
E.g. $(-16 + 4\mu)\mathbf{i} + (32 - 3\mu)\mathbf{j} + (10 + 5\mu)\mathbf{k} = 4\lambda \mathbf{j} + 6\lambda \mathbf{k} \Rightarrow -16 + 4\mu = 0 \Rightarrow \mu = \dots$

May also solve simultaneously using y and z components to find λ or μ

A1: $|\overrightarrow{OD}| = 10\sqrt{13}$

Note that the correct vector is $20\mathbf{j} + 30\mathbf{k}$

(b) Similar triangle approach



M1: For the key step in recognising that triangle BCD and triangle BAO are similar with a ratio of lengths of 4:1

dM1: States or uses the fact that $|\overrightarrow{OD}| = 5 \times |\overrightarrow{OB}|$

Stating this will score M1 dM1 provided there is no evidence of incorrect work

Note that they may establish this result using the work from (a) but must be used here to score.

A1: $|\overrightarrow{OD}| = 10\sqrt{13}$

Q4.

Question Number	Scheme	Marks	AO's
	Attempts any one of $(\pm \overrightarrow{PQ} =) \pm(\mathbf{q} - \mathbf{p}), (\pm \overrightarrow{PR} =) \pm(\mathbf{r} - \mathbf{p}), (\pm \overrightarrow{QR} =) \pm(\mathbf{r} - \mathbf{q})$ Or e.g. $(\pm \overrightarrow{PQ} =) \pm(\overrightarrow{OQ} - \overrightarrow{OP}), (\pm \overrightarrow{PR} =) \pm(\overrightarrow{OR} - \overrightarrow{OP}), (\pm \overrightarrow{QR} =) \pm(\overrightarrow{OR} - \overrightarrow{OQ})$	M1	1.1b
	Attempts e.g. $\mathbf{r} - \mathbf{q} = 2(\mathbf{q} - \mathbf{p})$ $\mathbf{r} - \mathbf{p} = 3(\mathbf{q} - \mathbf{p})$ $\frac{2}{3}(\mathbf{q} - \mathbf{p}) = \frac{1}{3}(\mathbf{r} - \mathbf{q})$ $\mathbf{q} = \mathbf{p} + \frac{1}{3}(\mathbf{r} - \mathbf{p})$ $\mathbf{q} = \mathbf{r} + \frac{2}{3}(\mathbf{p} - \mathbf{r})$	dM1	3.1a
	E.g. $\Rightarrow \mathbf{r} - \mathbf{q} = 2\mathbf{q} - 2\mathbf{p} \Rightarrow 2\mathbf{p} + \mathbf{r} = 3\mathbf{q} \Rightarrow \mathbf{q} = \frac{1}{3}(\mathbf{r} + 2\mathbf{p})^*$	A1*	2.1
		(3)	
(3 marks)			

Notes:

M1: Attempts any of the relevant vectors by subtracting either way around. This may be implied by sight of any one of $\pm(\mathbf{q} - \mathbf{p})$, $\pm(\mathbf{r} - \mathbf{p})$, $\pm(\mathbf{r} - \mathbf{q})$ ignoring how they are labelled

dM1: Uses the given information and writes it correctly in vector form that if rearranged would give the printed answer

A1*: Fully correct work leading to the given answer. Allow $\mathbf{OQ} = \dots$ as long as $\mathbf{OQ}$ has been defined as $\mathbf{q}$ earlier.

In the working allow use of $\mathbf{P}$ instead of $\mathbf{p}$ and $\mathbf{Q}$ instead of $\mathbf{q}$ as long as the intention is clear.

Q5.

Question	Scheme	Marks	AOs
	Attempts $\vec{AC} = \vec{AB} + \vec{BC} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k} + \mathbf{i} - 9\mathbf{j} + 3\mathbf{k} = 3\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}$	M1	3.1a
	Attempts to find any one length using 3-d Pythagoras	M1	2.1
	Finds all of $ AB = \sqrt{14}$, $ AC = \sqrt{61}$, $ BC = \sqrt{91}$	A1ft	1.1b
	$\cos BAC = \frac{14 + 61 - 91}{2\sqrt{14}\sqrt{61}}$	M1	2.1
	angle $BAC = 105.9^\circ$ *	A1*	1.1b
		(5)	
(5 marks)			
Notes:			
M1: Attempts to find $\vec{AC}$ by using $\vec{AC} = \vec{AB} + \vec{BC}$ M1: Attempts to find any one length by use of Pythagoras' Theorem A1ft: Finds all three lengths in the triangle. Follow through on their $ AC $ M1: Attempts to find BAC using $\cos BAC = \frac{ AB ^2 + AC ^2 - BC ^2}{2 AB AC }$ Allow this to be scored for other methods such as $\cos BAC = \frac{\vec{AB} \cdot \vec{AC}}{ AB AC }$ A1*: This is a show that and all aspects must be correct. Angle $BAC = 105.9^\circ$			

Q6.

Question	Scheme	Marks	AOs
(a)	$\overrightarrow{AB} = (3i - 3j - 4k) - (2i + 5j - 6k)$	M1	1.1b
	$= i - 8j + 2k$	A1	1.1b
		(2)	
(b)	States $\overrightarrow{OC} = 2 \times \overrightarrow{AB}$	M1	1.1b
	Explains that as OC is parallel to AB , so $OABC$ is a trapezium.	A1	2.4
		(2)	
(4 marks)			
Notes:			

(a)

M1: Attempts to subtract either way around. If no method is seen it is implied by two of $\pm i \pm 8j \pm 2k$.

A1: $i - 8j + 2k$ or $\begin{pmatrix} 1 \\ -8 \\ 2 \end{pmatrix}$ but not $(1, -8, 2)$

(b)

M1: Compares their $i - 8j + 2k$ with $2i - 16j + 4k$ by stating any one of

- $\overrightarrow{OC} = 2 \times \overrightarrow{AB}$
- $\begin{pmatrix} 2 \\ -16 \\ 4 \end{pmatrix} = 2 \times \begin{pmatrix} 1 \\ -8 \\ 2 \end{pmatrix}$
- $\overrightarrow{OC} = \lambda \times \overrightarrow{AB}$ or vice versa

This may be awarded if AB was subtracted "the wrong way around" or if there was one numerical slip

A1: A full explanation as to why $OABC$ is a trapezium.

Requires fully correct calculations, so part (a) must be $\overrightarrow{AB} = (i - 8j + 2k)$

It requires a reason and minimal conclusion.

Example 1:

$\overrightarrow{OC} = 2 \times \overrightarrow{AB}$, therefore OC is parallel to AB so $OABC$ is a trapezium

Example 2:

A trapezium has one pair of parallel sides. As $\overrightarrow{OC} = 2 \times \overrightarrow{AB}$, they are parallel, so ✓.

Example 3

As $\begin{pmatrix} 2 \\ -16 \\ 4 \end{pmatrix} = 2 \times \begin{pmatrix} 1 \\ -8 \\ 2 \end{pmatrix}$, OC and AB are parallel, so proven

Example 4

Accept as $\overrightarrow{OC} = \lambda \times \overrightarrow{AB}$, they are parallel so true

Note: There are two definitions for a trapezium. One stating that it is a shape with one pair of parallel sides, the other with **only one** pair of parallel sides. Any calculations to do with sides OA and CB in this question may be ignored, even if incorrect.

Q7.

Question	Scheme	Marks	AOs
(a)	Deduces that gradient of PA is $-\frac{1}{2}$	M1	2.2a
	Finding the equation of a line with gradient $-\frac{1}{2}$ and point $(7,5)$ $y - 5 = -\frac{1}{2}(x - 7)$	M1	1.1b
	Completes proof $2y + x = 17$ *	A1*	1.1b
		(3)	
(b)	Solves $2y + x = 17$ and $y = 2x + 1$ simultaneously	M1	2.1
	$P = (3, 7)$	A1	1.1b
	Length $PA = \sqrt{(3-7)^2 + (7-5)^2} = (\sqrt{20})$	M1	1.1b
	Equation of C is $(x-7)^2 + (y-5)^2 = 20$	A1	1.1b
		(4)	
(c)	Attempts to find where $y = 2x + k$ meets C using $\overrightarrow{OA} + \overrightarrow{PA}$	M1	3.1a
	Substitutes their $(11, 3)$ in $y = 2x + k$ to find k	M1	2.1
	$k = -19$	A1	1.1b
		(3)	
(10 marks)			

(c)	Attempts to find where $y = 2x + k$ meets C via simultaneous equations proceeding to a 3TQ in x (or y) FYI $5x^2 + (4k - 34)x + k^2 - 10k + 54 = 0$	M1	3.1a
	Uses $b^2 - 4ac = 0$ oe and proceeds to $k = \dots$	M1	2.1
	$k = -19$	A1	1.1b
		(3)	

Notes:

(a)

M1: Uses the idea of perpendicular gradients to deduce that gradient of PA is $-\frac{1}{2}$. Condone $-\frac{1}{2}x$ if followed by correct work. You may well see the perpendicular line set up as $y = -\frac{1}{2}x + c$ which scored this mark

M1: Award for the method of finding the equation of a line with a changed gradient and the point $(7, 5)$

So sight of $y - 5 = \frac{1}{2}(x - 7)$ would score this mark

If the form $y = mx + c$ is used expect the candidates to proceed as far as $c = \dots$ to score this mark.

A1*: Completes proof with no errors or omissions $2y + x = 17$

(b)

M1: Awarded for an attempt at the key step of finding the coordinates of point P , ie for an attempt at solving $2y + x = 17$ and $y = 2x + 1$ simultaneously. Allow any methods (including use of a calculator) but it must be a valid attempt to find both coordinates. Do not allow where they start $17 - x = 2x + 1$ as they have set $2y = y$ but condone bracketing errors, eg $2 \times 2x + 1 + x = 17$

A1: $P = (3, 7)$

M1: Uses Pythagoras' Theorem to find the radius or radius ² using their $P = (3, 7)$ and $(7, 5)$. There must be an attempt to find the difference between the coordinates in the use of Pythagoras

A1: $(x-7)^2 + (y-5)^2 = 20$. Do not accept $(x-7)^2 + (y-5)^2 = (\sqrt{20})^2$

(c)

M1: Attempts to find where $y = 2x + k$ meets C .

Awarded for using $\overrightarrow{OA} + \overrightarrow{PA}$. $(11, 3)$ or one correct coordinate of $(11, 3)$ is evidence of this award.

M1: For a full method leading to k . Scored for either substituting their $(11, 3)$ in $y = 2x + k$

or, in the alternative, for solving their $(4k-34)^2 - 4 \times 5 \times (k^2 - 10k + 54) = 0 \Rightarrow k = \dots$ Allow use of a calculator here to find roots. Award if you see use of correct formula but it would be implied by $\pm$ correct roots

A1: $k = -19$ only

Alternative I

M1: For solving $y = 2x + k$ with their $(x-7)^2 + (y-5)^2 = 20$ and creating a quadratic eqn of the form $ax^2 + bx + c = 0$ where both b and c are dependent upon k . The terms in x^2 and x must be collected together or implied to have been collected by their correct use in " $b^2 - 4ac$ "

FYI the correct quadratic is $5x^2 + (4k-34)x + k^2 - 10k + 54 = 0$

M1: For using the discriminant condition $b^2 - 4ac = 0$ to find k . It is not dependent upon the previous M and may be awarded from only one term in k .

$(4k-34)^2 - 4 \times 5 \times (k^2 - 10k + 54) = 0 \Rightarrow k = \dots$ Allow use of a calculator here to find roots.

Award if you see use of correct formula but it would be implied by $\pm$ correct roots

A1: $k = -19$ only

Alternative II

M1: For solving $2y + x = 17$ with their $(x-7)^2 + (y-5)^2 = 20$, creating a 3TQ and solving.

M1: For substituting their $(11, 3)$ into $y = 2x + k$ and finding k

A1: $k = -19$ only

.....
Other method are possible using trigonometry.